Focusing Heart Health of Old Age People through Experimental Analysis using Explainable AI for Personalized Assistive System *

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Abstract

The assistive systems in healthcare play a crucial role in improving the diagnosis of diseases. Personalization of health diagnostics is very important in tracking significant signs related to health issues. This study focuses on identifying key attributes linked to cardiac health in elderly people using machine learning and explainable AI techniques. Various machine learning models are used to predict cardiovascular disease, and their performance is evaluated. In addition, a diverse dataset of elderly patients is utilized to ensure robust validation of the developed prediction model. Explainable AI methods, such as SHapley Additive exPlanations, help to select relevant features that capture age-related cardiovascular risk factors. The proposed framework aims to support cardiologists in making informed treatment decisions for patients with elderly heart disease.

Keywords: Heart disease, Assistive System, Explainable AI

1 Introduction

Cardiovascular diseases (CVDs) are a major global health issue, particularly among the aging population. As the number of elderly individuals rises, so does the incidence of CVDs, highlighting the need for accurate and timely diagnosis. Early identification and risk evaluation are essential to prevent serious cardiac complications and enhance the quality of life for older adults. However, traditional assessment methods often fall short in predicting CVDs effectively in this demographic. Consequently, there is a growing need for advanced and reliable solutions to address this challenge.

Artificial Intelligence (AI) has shown great potential in transforming healthcare, including cardiovascular disease prediction. AI excels in processing large and complex datasets, detecting patterns, and building predictive models. These strengths make it a valuable tool for assessing cardiovascular risks in the elderly. By analyzing diverse patient data, such as medical records, imaging, genetic markers, and lifestyle habits, AI can generate personalized risk predictions. This leads to improved clinical decision-making and more targeted preventive strategies.

This paper explores key factors influencing cardiovascular disease in elderly individuals using explainable AI. The unique characteristics of older adults, such as multiple health conditions and physiological changes, can impact the accuracy of existing prediction models. Our main goals are:

1. To evaluate the effectiveness of different Machine Learning (ML) models in predicting CVDs among elderly individuals.

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2. To validate the developed prediction model using a diverse dataset representing the elderly population.
3. To identify and integrate age-specific cardiovascular risk factors using explainable AI techniques.

The paper is structured as follows: Section 2 reviews previous research on cardiovascular disease prediction, particularly in elderly populations. Section 3 describes the proposed approach. Section 4 details the experimental setup, datasets, and model evaluation results. Finally, Section 5 summarizes the findings, emphasizing the role of AI in elderly CVD prediction and its implications for personalized healthcare strategies.

2 Related Work

Several studies have explored cardiology using diverse datasets and methods to address various related challenges.

In [1], researchers analyzed the Cleveland Dataset from the UCI repository to predict heart disease. With 303 instances, data partitioning was done using K-fold Cross Validation. Six classifiers, including Decision Tree, Naïve Bayes, Multilayer Perceptron, KNN, Radial Basis Function, and SVM, were employed. Additionally, an ensemble approach utilizing bagging, boosting, and stacking was tested. The highest accuracy, 86.13%, was obtained using an ensemble SVM model with the dagging technique. The limited dataset size prompted further enhancement strategies for accuracy improvement.

Using the same dataset, N. Louridi et al. [2] emphasized data preprocessing for better predictive performance. Missing values were replaced with the mean, and classifiers like SVM, KNN, and Naïve Bayes were applied. The study concluded that SVM with a linear kernel yielded an accuracy of 86.8%.

S. Islam et al. [3] utilized Logistic Regression, Naïve Bayes, SVM, and Decision Tree models for heart disease prediction. Performance was assessed via confusion matrices, revealing that Logistic Regression, using 12 attributes, achieved 86.25% accuracy. Researchers suggested expanding the dataset to further enhance model accuracy.

In [4], the authors leveraged the CVD and Framingham datasets to assess feature correlation and selection's role in predictive accuracy. Anova, a filter-based feature selection technique, was applied to determine significant feature relationships. Key risk factors identified included age, hypertension, glucose levels, past heart conditions, and blood pressure. The reduced feature set improved prediction accuracy from 0.73 to 0.74 for the CVD dataset and from 0.66 to 0.71 for the Framingham dataset.

N. G. Bhuvaneswari Amma [5] developed an intelligent system combining genetic algorithms and neural networks for heart disease prediction. Data preprocessing included normalization of age, blood sugar, cholesterol, heart rate, and ST depression. Using a Backpropagation Algorithm, the model classified disease severity into five categories, achieving 94.17% accuracy. The study suggested Principal Component Analysis for dimensionality reduction to enhance performance.

A decision support system proposed in [6] applied SVM and MLP neural networks on the UCI dataset. The SVM model classified heart disease presence or absence with 80.14% accuracy, while the ANN model with 13 nodes categorized patients into five risk levels, achieving 97.55% accuracy.

Pahulpreet Singh Kohli and Shriya Arora [7] explored machine learning applications in disease prediction. Algorithms such as Logistic Regression, Decision Trees, Random Forest, SVM, and Adaptive Boosting were employed. Logistic Regression delivered the highest accuracy (87.1%), with automation of data processing steps suggested for improvement.

Advait Shirvaikar et al. [8] utilized a Kaggle dataset containing 70,000 records and 10 features. After preprocessing, an additional BMI feature was derived from weight and height, showing strong correlation with disease risk. Feature selection based on correlation thresholding led to testing with KNN, Random Forest, and SVM. The Random Forest model achieved the best accuracy of 73%.

Aditi Gavhane et al. [9] investigated the Extreme Learning Machine (ELM) approach for heart disease diagnosis. The proposed ELM model, configured with 100 neurons, achieved an accuracy of 86.5% and provided a disease presence score on a 0-4 scale. The study highlighted the potential of estimating attribute contributions to mitigate missing data issues.

In [10], researchers analyzed SVM, Random Forest, XGBoost, and KNN models on the Cleveland Heart Disease Dataset. Instead of prioritizing accuracy, they focused on interpretability using SHAP and LIME. The top features identified were ca, cp, thal, thalach, and oldpeak. However, the role of age as a key feature remained inconclusive.

In [11], a study on a 70,000-patient dataset employed LASSO feature selection and ensemble learning techniques. Bagging models improved average accuracy by 1.96%, boosting produced the highest AUC score of 0.73, and a stacked model combining KNN, SVM, and Random Forest achieved 75.1% accuracy.

In [12], ten machine learning classifiers were tested on the Cleveland dataset, using both full and optimized attribute sets. Cross-validation and hyperparameter tuning led to SMO achieving the highest accuracy—85.148% on the full set and 86.468% on a reduced set selected via chi-square evaluation.

3 Proposed Assistive System

This study develops a framework integrating the machine learning models with explainable AI using Heart failure prediction and Heart UCI datasets for heart disease prediction. The methodology includes four phases: data preprocessing, exploratory analysis, feature selection, and predictive modeling.

Categorical data, including demographic details, were numerically encoded, while numerical features underwent standardization using Min-Max scaling. Outliers were identified and removed via the interquartile range (IQR) method.

Exploratory data analysis (EDA) provided insights into dataset distribution. Visualizations, such as histograms and boxplots, highlighted variable relationships and imbalances. Chi-squared tests and ANOVA F-tests were used for categorical and numerical feature selection, respectively.

A weighted sum model quantified cardiovascular risk:

$$R = w_1 \cdot A + w_2 \cdot B + w_3 \cdot C + \cdots + w_n \cdot X \tag{1}$$

where R represents overall risk, w_i denotes feature weights, and X_i corresponds to feature values.

Five machine learning models were evaluated: Logistic Regression, Decision Trees, SVM, Random Forest, and KNN. Hyperparameter tuning via grid search optimized performance, with accuracy, ROC-AUC, and confusion matrices as evaluation metrics. Random Forest achieved the highest accuracy (89.3%), while SVM demonstrated superior sensitivity in early disease detection.

An ensemble model combining Logistic Regression, Random Forest, and SVM improved performance, achieving a 92.1% ROC-AUC score. SHAP analysis revealed that ST_Slope and Cholesterol levels were key predictors:

$$f(x) = \phi_0 + \sum_{i=1}^{n} \phi_i x_i \tag{2}$$

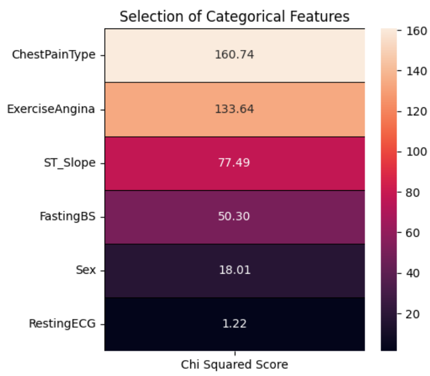


Figure 1: Feature Selection for Categorical Features Using Chi-Squared Test

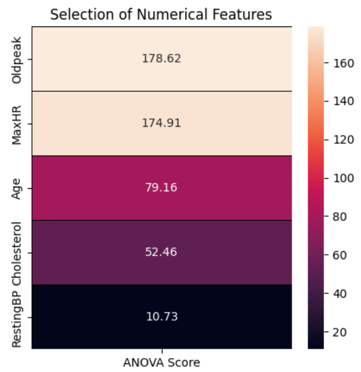


Figure 2: Feature Selection for Categorical Features Using ANOVA Scores

These findings support AI-driven early heart disease detection, highlighting its clinical relevance and potential for real-time deployment in healthcare settings.

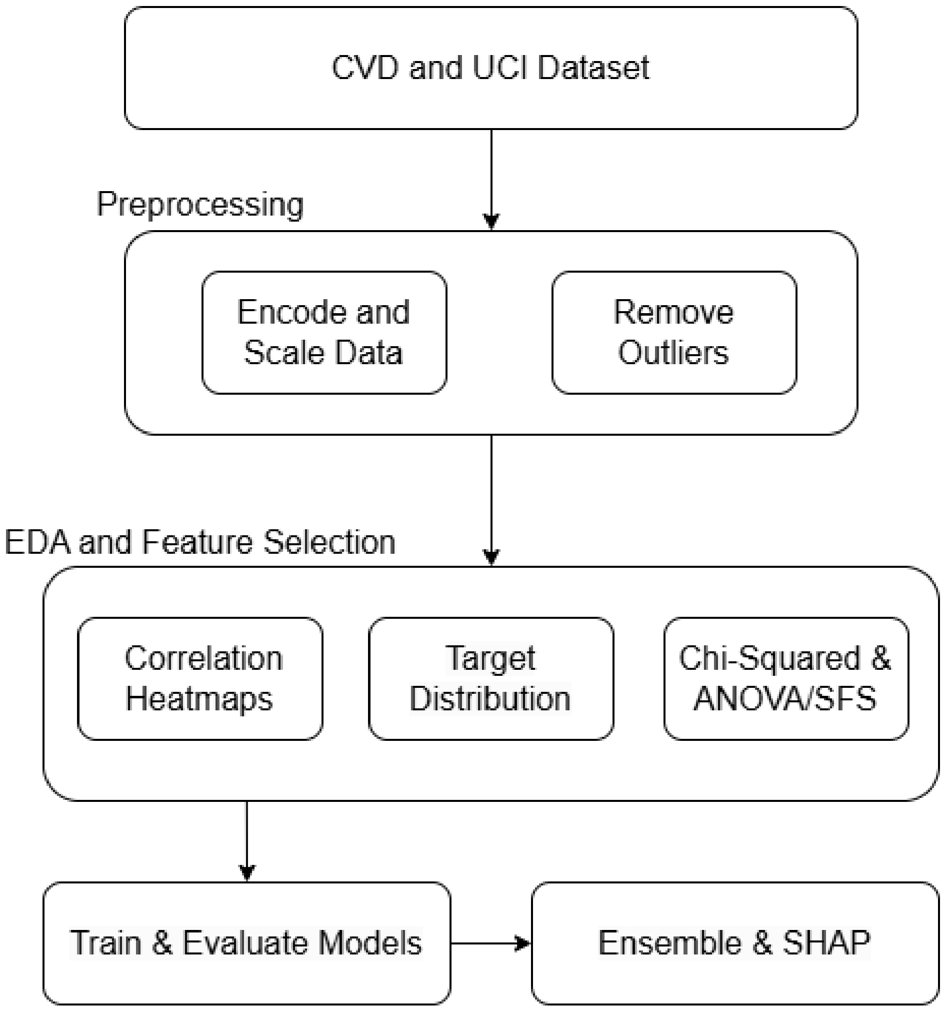


Figure 3: Proposed Methodology

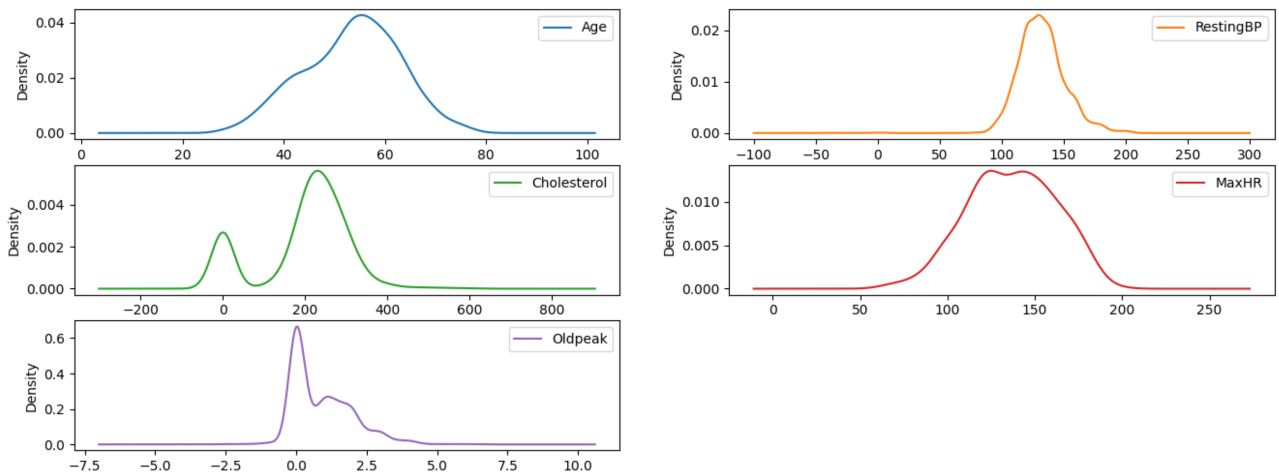


Figure 4: Density Plots for Numeric Features in the Heart Disease Dataset

4 Results and discussion

4.1 Datasets

4.1.1 UCI Heart Disease Dataset

The Cleveland Heart Disease dataset is a widely used multivariate dataset with 14 attributes designed to predict heart disease presence. It includes variables such as age, sex, chest pain type, blood pressure, cholesterol levels, and exercise-induced angina. This dataset supports coronary artery disease diagnosis and helps in recognizing heart condition patterns. Originally compiled from medical institutions like the Hungarian Institute of Cardiology and the Cleveland Clinic Foundation, it has played a crucial role in developing predictive heart disease models [13].

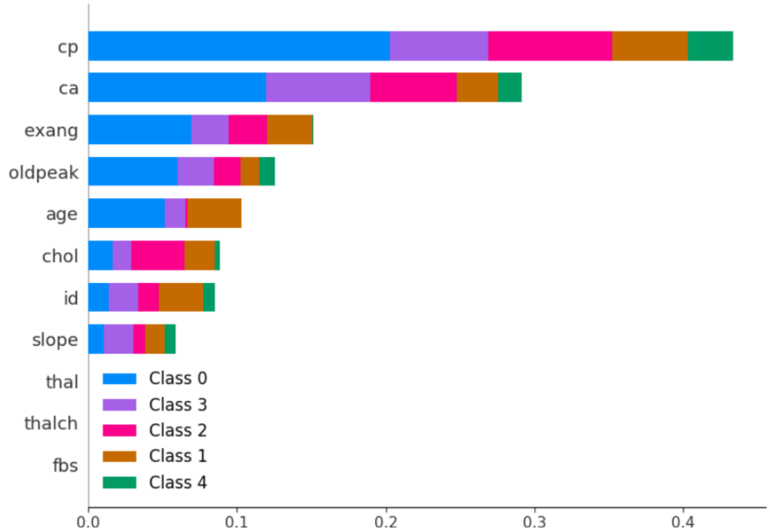
4.1.2 Heart Failure Prediction Dataset

The Heart Failure Prediction Dataset consists of 11 features useful in predicting heart disease, a major cause of mortality worldwide. This dataset was created by integrating data from five independent sources: Cleveland, Hungarian, Switzerland, Long Beach VA, and Stalog, resulting in 918 observations. It facilitates early cardiovascular disease detection and management through machine learning techniques. Available via the UCI Machine Learning Repository, it has been extensively used for research in predictive modeling for heart disease [15].

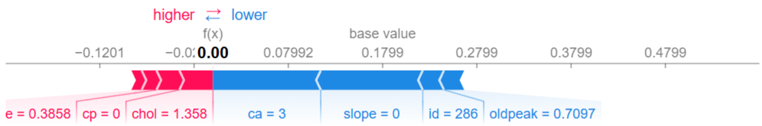
4.2 Observations

The Random Forest model exhibited the highest accuracy of 89.13%, surpassing other models. Its strong performance stems from its ability to handle complex feature interactions while maintaining robustness against overfitting, as presented in Table 1. The Support Vector Classifier (SVC) followed closely with an accuracy of 88.59%, benefiting from its effectiveness in high-dimensional spaces. Logistic Regression, despite its simplicity, provided consistent accuracy at 85.87%, showing its efficiency for linearly separable data. Decision Trees achieved an accuracy of 87.50%, demonstrating their capability to capture non-linear relationships.

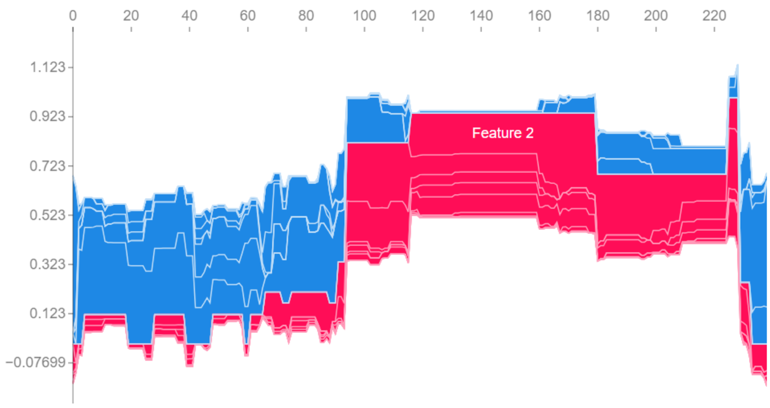
Feature importance analysis using SHapley Additive exPlanations (SHAP) identified critical predictors influencing heart disease risk. Factors such as age, cholesterol levels, and maximum heart rate emerged as key indicators across multiple models. The SHAP visualizations further illustrate these contributions, enhancing interpretability and transparency in decision-making.



(a) Feature importance summary plot with key factors like cholesterol and age



(b) SHAP force plot highlighting the impact of exercise and resting ECG



(c) SHAP force plot showing contributions of variables like age and exercise angina

Figure 5: SHAP analysis demonstrating feature importance in heart disease prediction through various plots.

5 Conclusion

This study underscores the significance of machine learning models, coupled with explainable AI, in supporting cardiovascular disease diagnosis among elderly patients. The Random Forest classifier demonstrated the highest accuracy (89.13%), followed by SVM (88.59%), Decision Trees (87.50%), and Logistic Regression (85.87%). Additionally, SHAP analysis provided valuable interpretability, pinpointing key features like cholesterol and maximum heart rate, older age, impact of exercise, thus increasing the models' reliability in clinical practice.

Future work could focus on integrating additional datasets, including genetic and lifestyle factors, to en-

Table 1: Hyperparameter Tuning Results

(a) Random Forest

Max Leaf	Min Samples Estimators	N	Accuracy	Depth
7	70	50	88.04	
7	50	50	86.4	
7	100	50	85.33	
5	70	50	85.59	
6	70	50	88.04	
6	70	40	88.59	
6	70	60	89.13	
6	70	70	88.59	

(b) Logistic Regression

C Value	Accuracy
0.1	83.43
0.5	84.78
1	85.33
5	85.87
10	85.87
20	85.87

(c) SVM

Kernel	C	Gamma	Accuracy
Linear	0.1	0	83.15
Linear	0.1	0.1	86.41
Linear	1	0.1	87.5
Linear	10	0.1	87.5
RBF	1	0.1	86.4
RBF	10	0.1	85.87
RBF	10	0.01	87.5

(d) Decision Tree

Max Depth	Min Leaf Samples	Accuracy
5	2	86.96
10	10	80.98
5	2	86.96
1	2	81.52
2	2	75.00
3	2	84.78
4	2	87.50

hance predictive accuracy. The exploration of deep learning methods and hybrid models may further optimize performance. Additionally, incorporating real-time health monitoring through wearable devices could facilitate continuous assessment and early intervention. Efforts to enhance model explainability will ensure the practical applicability of AI-driven tools in cardiology, aiding healthcare professionals in making well-informed treatment decisions.

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